

# **Companion to Bird & de Moor, Algebra of Programming**

## §6. Recursive Programs

### §6.1. Digits of a number

(6.1a)

$$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \text{ } \neg\text{Digit}(\text{val}^\circ) \text{ snoc})$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

the converse of  $\text{val}$ , which reads digits as a number

(a)
(b)
(c)
(d)
(e)

|     |                                                                                                                          |                                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{val}^\circ$                                                                                                       | $\text{val}$ undoes $\text{wrap}$ or $\text{snoc}$ , reads the front with $\text{val}$ and combines by $\text{embed}$ or $\text{op}$ , so its converse runs that backwards |
| (b) | $= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-} (\neg\text{Digit}(\text{val}))) [\text{embed}, \text{op}]^\circ$ | undoing $\text{wrap}$ or $\text{snoc}$ first gives two cases: a decimal and its last digit read by $\text{val}$ then $\text{op}$ , or one digit read by $\text{embed}$     |
| (c) | $= (\text{wrap}^\circ \text{ embed})^\circ \cup (\text{snoc}^\circ \neg\text{Digit}(\text{val}) \text{ op})^\circ$       | the converse of each case is its steps undone in reverse: undo $\text{op}$ , undo reading the front, then $\text{snoc}$ ; or undo $\text{embed}$ , then $\text{wrap}$      |
| (d) | $= (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg\text{Digit}(\text{val})^\circ \text{ snoc})$             | undoing the read of the front while keeping the last digit is $\text{val}^\circ$ on the front alone                                                                        |
| (e) | $= (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg\text{Digit}(\text{val}^\circ) \text{ snoc})$             |                                                                                                                                                                            |

(6.1b)

- 1  $\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$   
 $\text{op}^\circ$  splits a number into its quotient and remainder by 10
- 2  $(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$   
 $\text{op}^\circ$  gives a pair with a nonzero first component exactly at the numbers with two or more digits
- 3  $(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$   
 $\text{embed}^\circ$  gives a digit exactly at the one-digit numbers

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$  .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$  .

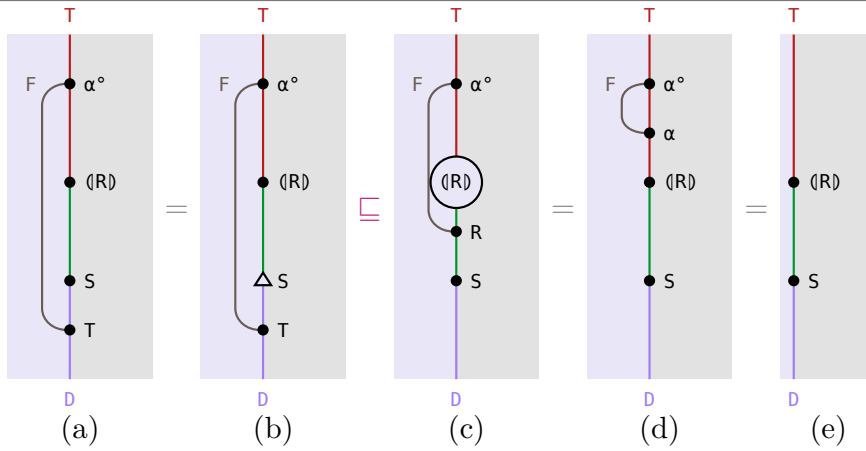
(6.2b)

| the law                                                                                   | what it says                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| 2 $\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| 3 $\alpha^\circ F(X) R \sqsubseteq X \implies \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| 4 $X \sqsubseteq \alpha^\circ F(X) R \implies X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

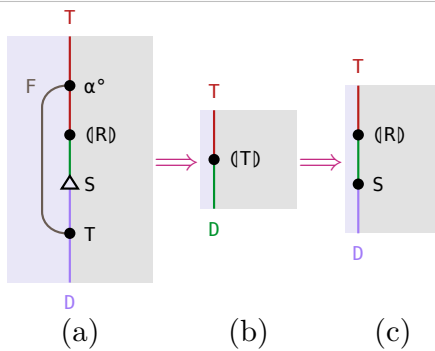
$$F(S) \ T \sqsubseteq R \ S \implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$



|     |                                                       |                                                         |
|-----|-------------------------------------------------------|---------------------------------------------------------|
| (a) | $\alpha^\circ F(\langle R \rangle S) T$               | $t(f \ g) = t(f) \ t(g)$                                |
| (b) | $= \alpha^\circ F(\langle R \rangle) F(S) T$          | $F(S) \ T \sqsubseteq R \ S$                            |
| (c) | $\sqsubseteq \alpha^\circ F(\langle R \rangle) R \ S$ | $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ |
| (d) | $= \alpha^\circ \alpha \ \langle R \rangle \ S$       | $\alpha^\circ \ \alpha = 1$                             |
| (e) | $= \langle R \rangle \ S$                             |                                                         |



|     |                                                                                    |                                                                                                                    |
|-----|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| (a) | $F(S) \ T \sqsubseteq R \ S$                                                       | $F(S) \ T \sqsubseteq R \ S$<br>$\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle \ S$ |
| (b) | $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle \ S$ | $\alpha^\circ F(X) R \sqsubseteq X$<br>$\implies \langle R \rangle \sqsubseteq X$                                  |
| (c) | $\implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$                     |                                                                                                                    |

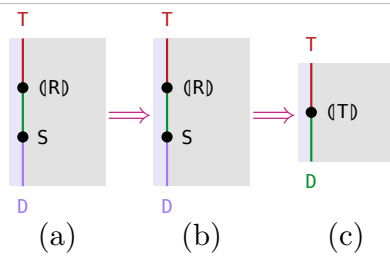
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|     |                                                           |                                                         |
|-----|-----------------------------------------------------------|---------------------------------------------------------|
| (a) | $\langle R \rangle \ S$                                   |                                                         |
| (b) | $= \langle R \rangle \ S$                                 | $\alpha^\circ \ \alpha = 1$                             |
| (c) | $= \alpha^\circ \ \alpha \ \langle R \rangle \ S$         | $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ |
| (d) | $= \alpha^\circ \ F(\langle R \rangle) \ R \ S$           | $R \ S \sqsubseteq F(S) \ T$                            |
| (e) | $\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ | $t(f \ g) = t(f) \ t(g)$                                |



|     |                                                                                             |                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| (a) | $R \ S \sqsubseteq F(S) \ T$                                                                | $R \ S \sqsubseteq F(S) \ T$<br>$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ |
| (b) | $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$ | $X \sqsubseteq \alpha^\circ \ F(X) \ R$<br>$\Rightarrow X \sqsubseteq \langle R \rangle$                                    |
| (c) | $\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$                           |                                                                                                                             |

## §6.3. Hylomorphisms

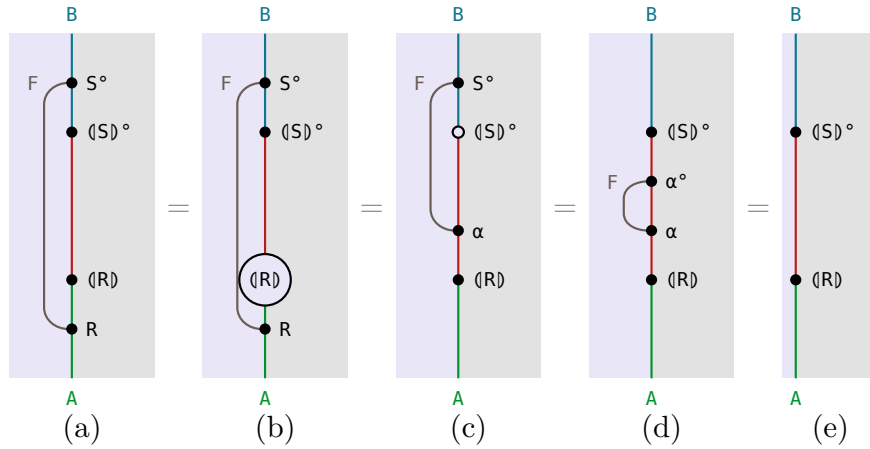
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R = (\mathbb{S})^\circ (\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $S^\circ$  represents the decomposition stage,  $F((\mathbb{S})^\circ (\mathbb{R}))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $S : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial

the body at  $(\mathbb{S})^\circ (\mathbb{R})$

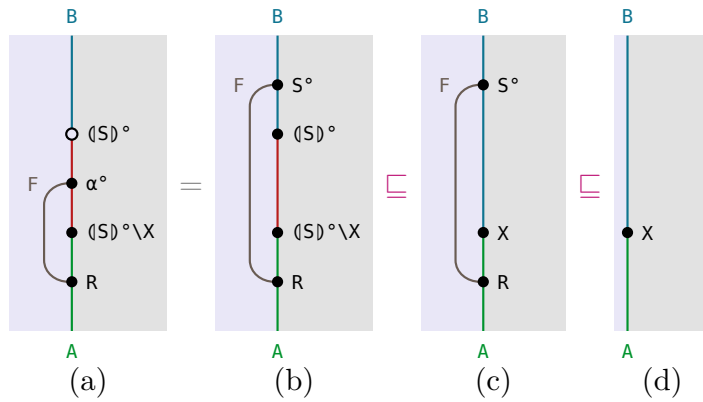


|     |                                                         |                                                                                                                 |
|-----|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| (a) | $S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R$          |                                                                                                                 |
| (b) | $= S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$     | $F((\mathbb{R}))R = \alpha(\mathbb{R}) \text{ — } \triangleq \text{reduce}$                                     |
| (c) | $= S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$   | $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \text{ — } \triangleq \text{reduce, (5.1b.3)}$ |
| (d) | $= (\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$ | $\alpha \text{ iso}$                                                                                            |
| (e) | $= (\mathbb{S})^\circ (\mathbb{R})$                     |                                                                                                                 |

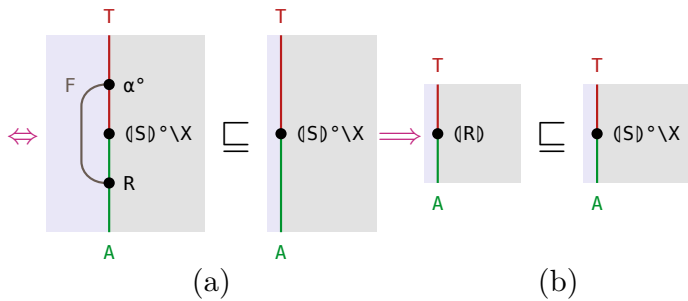
$$S^\circ F(X) R \sqsubseteq X \implies (\mathbb{S})^\circ (\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(\mathbb{S})^\circ (\mathbb{R})$  is included in  $X$  if  $X$  satisfies the associated recursion inequation

$Y := (\mathbb{S})^\circ \setminus X$



|     |                                                                       |                                                                                        |
|-----|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (a) | $(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$ | $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$                      |
| (b) | $= S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$  | $(\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \text{ — (0.9a.4)}$ |
| (c) | $\sqsubseteq S^\circ F(X) R$                                          | $S^\circ F(X) R \sqsubseteq X$                                                         |
| (d) | $\sqsubseteq X$                                                       | $S \cdot \neg S \text{ — (0.1a.10)}$                                                   |

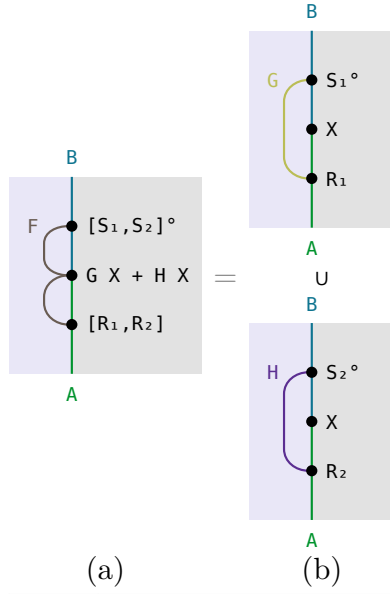


|     |                                                                                                                                                     |                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| (a) | $\Leftrightarrow S^\circ F(X) R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq (\mathbb{S})^\circ \setminus X$ | $(\mathbb{R}) = (\mu X : \alpha^\circ F(X) R) \text{ — } \triangleq \text{reduce, (6.2b.3), (0.1a.15)}$ |
| (b) | $\Rightarrow S^\circ F(X) R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$                                         |                                                                                                         |

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2))) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at  $F(X)=G(X)+H(X)$

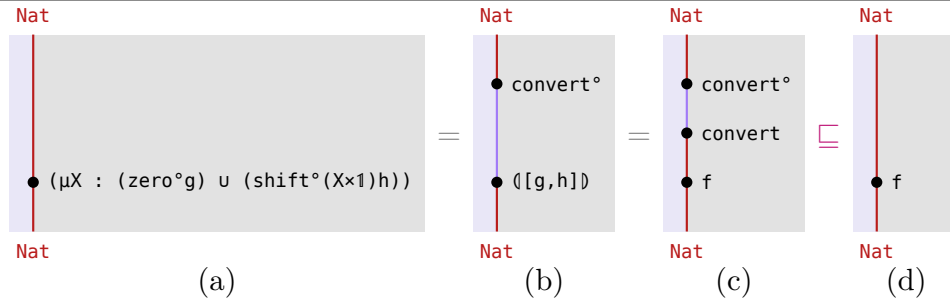


|     |                                                    |                                                                                                                                                                                              |
|-----|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$        | $[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$ ; the $+$ stands for the codiagonals $\nabla^\circ, \nabla$ , which are not drawn |
| (b) | $= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$ | $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct                                                                                                                                           |

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f \quad (6.4a)$$

once  $f$  sends **zero** to  $g$  and turns **shift** into  $h$ , the recursion that halves the argument at each step computes  $f$



|     |                                                                            |                                                                                                                                |
|-----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$ | $\text{convert}^\circ \llbracket [g, h] \rrbracket = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$ |
| (b) | $= \text{convert}^\circ \llbracket [g, h] \rrbracket$                      | $\text{zero } f = g \wedge \text{shift } f = (f \times 1) h$                                                                   |
| (c) | $= \text{convert}^\circ \text{convert } f$                                 | $\Rightarrow \text{convert } f = \llbracket [g, h] \rrbracket$                                                                 |
| (d) | $\sqsubseteq f$                                                            | $\text{Simple}(\text{convert})$                                                                                                |

$$(\mu X : (\text{zero}^\circ \text{one}) \cup (\text{shift}^\circ (X \times 1) \text{op}(a))) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes  $a$  to the power  $b$

$\text{zero } \text{exp}(a) = \text{one}$

$a$  to the power  $0$  is  $1$

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

$a$  to the power  $2n+d$  is  $\text{op}(a)$  of  $a$  to the power  $n$  and  $d$

$$(\mu X : (\text{zero}^\circ \text{zero}) \cup (\text{shift}^\circ (X \times 1) \text{op}(b))) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves  $a$  at each step computes the remainder of  $a$  by  $b$

$\text{zero } \text{mod}(b) = \text{zero}$

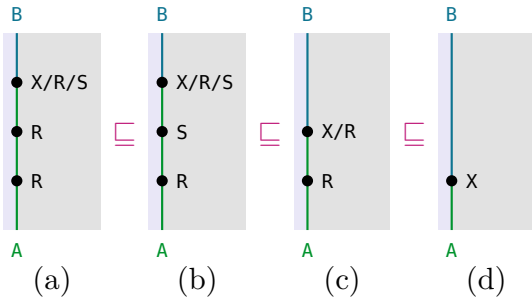
$0 \text{ mod } b$  is  $0$

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

$(2a+d) \text{ mod } b$  is  $\text{op}(b)$  of  $a \text{ mod } b$  and  $d$

## §6.5. Unique fixed points

$$\begin{aligned}
 & \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{X} & (6.5a) \\
 & ((\mathbf{X}/\mathbf{R})/\mathbf{S}) \text{ followed by } \mathbf{R}\mathbf{R} \text{ stays below } \mathbf{X} \text{ when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \\
 & \mathbf{inductive}(\mathbf{S}) \wedge \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow \mathbf{inductive}(\mathbf{R}) \\
 & \mathbf{R} \text{ is inductive when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \text{ for an inductive } \mathbf{S}
 \end{aligned}$$



|     |                                                                            |                                                               |
|-----|----------------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $(\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R}$             | $\mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R}$     |
| (b) | $\sqsubseteq (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{S} \mathbf{R}$ | $(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$ |
| (c) | $\sqsubseteq (\mathbf{X} / \mathbf{R}) \mathbf{R}$                         | $(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$ |
| (d) | $\sqsubseteq \mathbf{X}$                                                   |                                                               |

$$\mathbf{member} = \emptyset \quad (6.5b)$$

a constant relator records no elements, so its membership is empty

$$\mathbf{member} = [\mathbf{member}, \mathbf{member}]$$

a member of  $\mathbf{F}+\mathbf{G}$  is a member of whichever summand the value is in

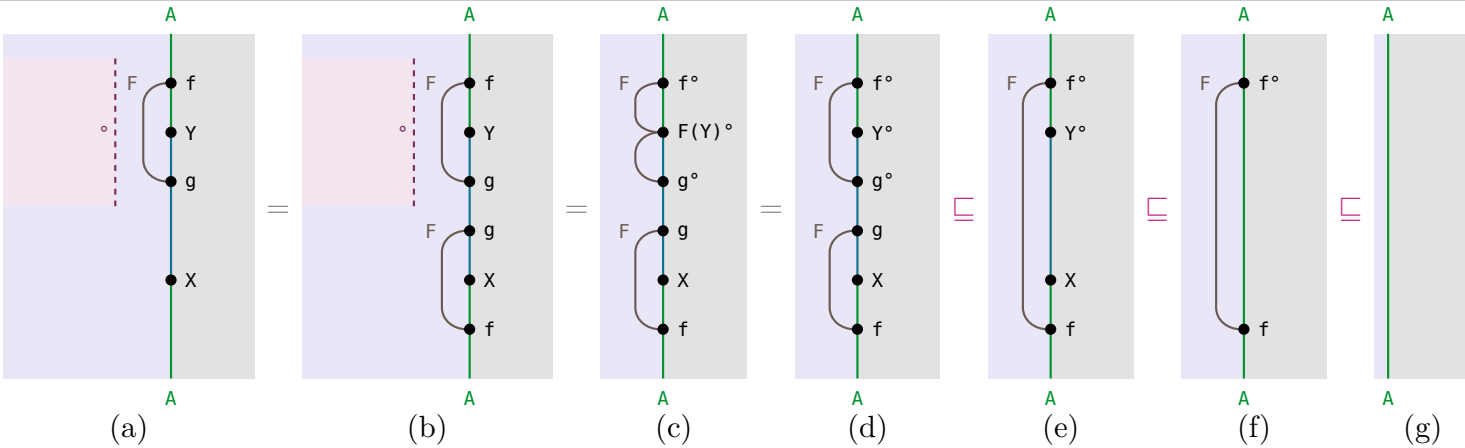
$$\mathbf{member} = (\pi_1 \mathbf{member}) \cup (\pi_2 \mathbf{member})$$

a member of  $\mathbf{F} \times \mathbf{G}$  is a member of either component

$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{X} = \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \wedge \mathbf{Y} = \mathbf{S} \mathbf{F}(\mathbf{Y}) \mathbf{R} \Rightarrow \mathbf{X} = \mathbf{Y} & (6.5c) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive, two solutions } \mathbf{X}, \mathbf{Y} \text{ of } \mathbf{X} = \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ are equal}
 \end{aligned}$$

$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{Entire}(\mathbf{S}) \wedge \mathbf{Entire}(\mathbf{R}) \wedge \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \sqsubseteq \mathbf{X} \Rightarrow \mathbf{Entire}(\mathbf{X}) & (6.5d) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive and } \mathbf{S}, \mathbf{R} \text{ are entire, a pre-fixed point of } \mathbf{X} \mapsto \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ is entire}
 \end{aligned}$$

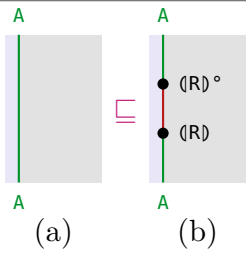
$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{g} \mathbf{member}) \wedge \mathbf{Map}(\mathbf{g}) \wedge \mathbf{Map}(\mathbf{f}) \wedge \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \Rightarrow \mathbf{Map}(\mathbf{X}) & (6.5e) \\
 & \text{when } \mathbf{g} \mathbf{member} \text{ is inductive and } \mathbf{f}, \mathbf{g} \text{ are maps, the solution of } \mathbf{X} = \mathbf{g}\mathbf{F}(\mathbf{X})\mathbf{f} \text{ is a map} \\
 & \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \wedge \mathbf{Simple}(\mathbf{g}) \wedge \mathbf{Simple}(\mathbf{f}) \wedge \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1} \Rightarrow (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X} \sqsubseteq \mathbf{1} \\
 & (\mathbf{g}\mathbf{F}(\mathbf{Y})\mathbf{f})^\circ \text{ followed by } \mathbf{X} \text{ stays below } \mathbf{1} \text{ when } \mathbf{g}, \mathbf{f} \text{ are simple and } \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}
 \end{aligned}$$



|     |                                                                                                                 |                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| (a) | $(\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X}$                                               | $\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$                         |
| (b) | $= (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$           | $(\mathbf{R} \mathbf{S})^\circ = \mathbf{S}^\circ \mathbf{R}^\circ$                 |
| (c) | $= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y})^\circ \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$ | $\mathbf{PreservesRecip}(\mathbf{F})$                                               |
| (d) | $= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ) \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$ | $\mathbf{Simple}(\mathbf{g})$                                                       |
| (e) | $\sqsubseteq \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ \mathbf{X}) \mathbf{f}$                               | $\mathbf{t}(\mathbf{f} \mathbf{g}) = \mathbf{t}(\mathbf{f}) \mathbf{t}(\mathbf{g})$ |
| (f) | $\sqsubseteq \mathbf{f}^\circ \mathbf{f}$                                                                       | $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$                                |
| (g) | $\sqsubseteq \mathbf{1}$                                                                                        | $\mathbf{t}(\mathbf{1}) = \mathbf{1}$                                               |

(6.5f)

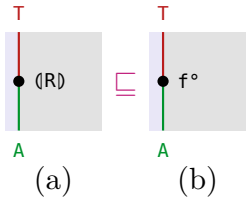
**inductive( $R^\circ$  member)  $\wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R)$**   
 when  $R^\circ$ member is inductive and  $R$  is surjective, the fold  $(R)$  is surjective



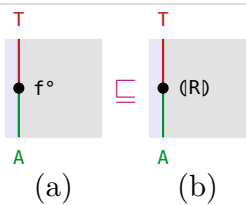
|     |                             |                       |
|-----|-----------------------------|-----------------------|
| (a) | $1$                         | (6.5d) at $S=R^\circ$ |
| (b) | $\sqsubseteq (R)^\circ (R)$ |                       |

(6.5g)

**Map( $f$ )  $\wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R)$**   
 a map  $f$  with  $Rf \sqsubseteq F(f) \alpha$ , for a surjective  $R$ , has the fold  $(R)$  as its converse



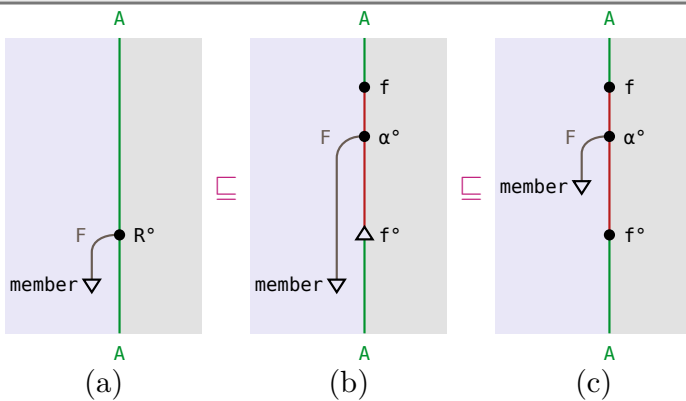
|     |                       |                                                     |
|-----|-----------------------|-----------------------------------------------------|
| (a) | $(R)$                 | shunting $f$ , fusion, $Rf \sqsubseteq F(f) \alpha$ |
| (b) | $\sqsubseteq f^\circ$ |                                                     |



|     |                   |                                  |
|-----|-------------------|----------------------------------|
| (a) | $f^\circ$         | (6.5f) and $(R) f \sqsubseteq 1$ |
| (b) | $\sqsubseteq (R)$ |                                  |

(6.5h)

**Map( $f$ )  $\wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ$  member  $\sqsubseteq f \alpha^\circ$  member  $f^\circ$**   
 $R^\circ$ member is below  $\alpha^\circ$ member conjugated by  $f$ , so it is inductive when  $\alpha^\circ$ member is



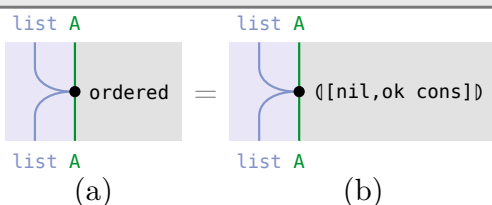
|     |                                                |                                                             |
|-----|------------------------------------------------|-------------------------------------------------------------|
| (a) | $R^\circ$ member                               | Map( $f$ ) $\wedge R f \sqsubseteq F(f) \alpha$             |
| (b) | $\sqsubseteq f \alpha^\circ F(f^\circ)$ member | $\Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$ |
| (c) | $\sqsubseteq f \alpha^\circ$ member $f^\circ$  | $F(R)$ member $\sqsubseteq$ member $R$                      |

## §6.6. Sorting by selection

(6.6a)

**ordered =  $([nil, ok \text{ cons}])$**

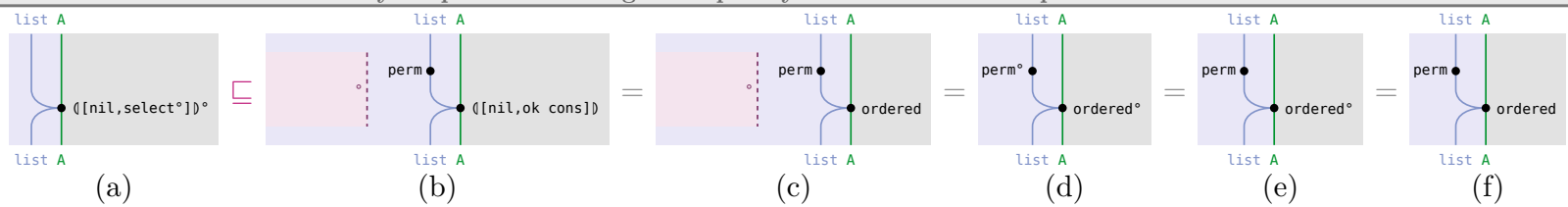
a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it



|     |                                                |                 |
|-----|------------------------------------------------|-----------------|
| (a) | <b>ordered</b>                                 | fold uniqueness |
| (b) | <b>= <math>([nil, ok \text{ cons}])</math></b> |                 |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

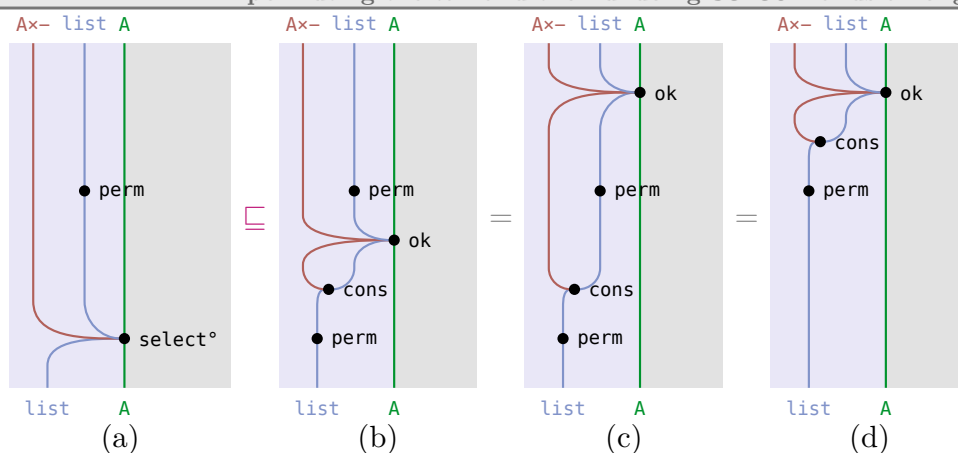
every output of unfolding the input by **select** is a sorted permutation of it



|     |                                                                      |                                                                                                                                                                                     |
|-----|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{select}^\circ]^\circ$                            | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$<br>$\Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{perm}^\circ$ |
| (b) | $\sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{perm}^\circ$ | (6.6a)                                                                                                                                                                              |
| (c) | $= (\text{ordered perm})^\circ$                                      | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                     |
| (d) | $= \text{perm}^\circ \text{ordered}^\circ$                           | $\text{perm}^\circ = \text{perm}$                                                                                                                                                   |
| (e) | $= \text{perm ordered}^\circ$                                        | $\text{ordered}^\circ = \text{ordered}$                                                                                                                                             |
| (f) | $= \text{perm ordered}$                                              |                                                                                                                                                                                     |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

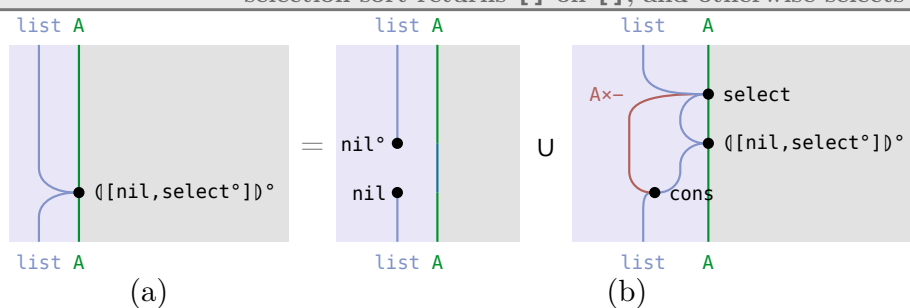
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



|     |                                                          |                                                                       |
|-----|----------------------------------------------------------|-----------------------------------------------------------------------|
| (a) | $(1 \times \text{perm}) \text{select}^\circ$             | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                 |
| (b) | $\sqsubseteq (1 \times \text{perm}) \text{ok cons perm}$ | $\text{ok} (1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$ |
| (c) | $= \text{ok} (1 \times \text{perm}) \text{cons perm}$    | $\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$          |
| (d) | $= \text{ok cons perm}$                                  |                                                                       |

$$([\text{nil}, \text{select}^\circ])^\circ = (\text{nil}^\circ \text{nil}) \cup (\text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons}) \quad (6.6d)$$

selection sort returns **[]** on **[]**, and otherwise selects **(a, y)**, sorts **y** and conses **a** back on

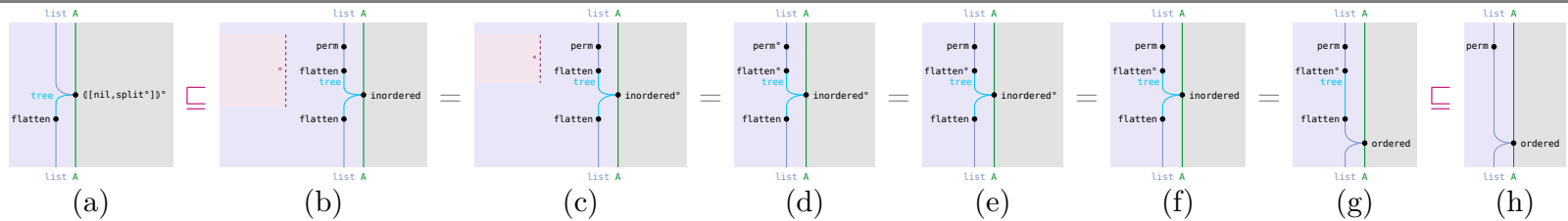


|     |                                                                                                                         |                                  |
|-----|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| (a) | $[\text{nil}, \text{select}^\circ]^\circ$                                                                               | unfolding the converse of a fold |
| (b) | $= (\text{nil}^\circ \text{nil}) \cup (\text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons})$ |                                  |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

**flatten**  $\sqsubseteq$  **perm ordered**

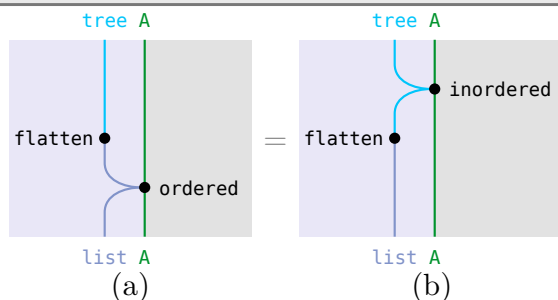
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



|     |                                                                                     |                                                                                                                                                                                                              |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{split}^\circ]^\circ \text{ flatten}$                            | $\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$<br>$\Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$ |
| (b) | $\sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$                 | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                                              |
| (c) | $= (\text{flatten perm})^\circ \text{ inordered}^\circ \text{ flatten}$             | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                                              |
| (d) | $= \text{perm}^\circ \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$ | $\text{perm}^\circ = \text{perm}$                                                                                                                                                                            |
| (e) | $= \text{perm flatten}^\circ \text{ inordered}^\circ \text{ flatten}$               | $\text{inordered}^\circ = \text{inordered}$                                                                                                                                                                  |
| (f) | $= \text{perm flatten}^\circ \text{ inordered flatten}$                             | (6.6f)                                                                                                                                                                                                       |
| (g) | $= \text{perm flatten}^\circ \text{ flatten ordered}$                               | $\text{flatten}^\circ \text{ flatten} \sqsubseteq 1$                                                                                                                                                         |
| (h) | $\sqsubseteq \text{perm ordered}$                                                   |                                                                                                                                                                                                              |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

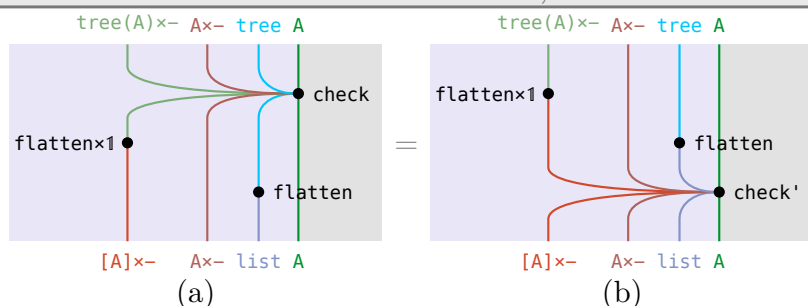
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



|     |                              |                                                                       |
|-----|------------------------------|-----------------------------------------------------------------------|
| (a) | <b>flatten ordered</b>       | R transitive: a flattened tree is ordered iff every node passes check |
| (b) | $= \text{inordered flatten}$ |                                                                       |

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6g)$$

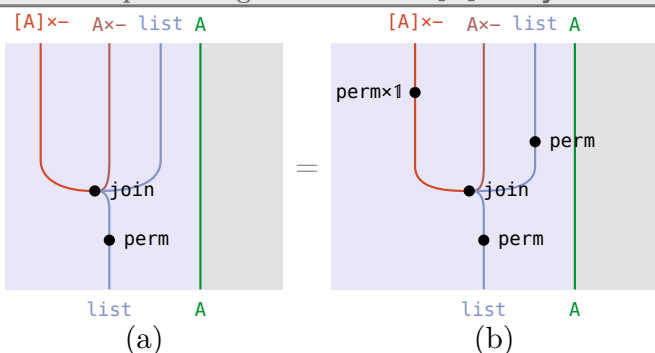
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



|     |                                                                      |                                                                |
|-----|----------------------------------------------------------------------|----------------------------------------------------------------|
| (a) | $\text{check} (\text{flatten} \times (1 \times \text{flatten}))$     | $b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$ |
| (b) | $= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$ |                                                                |

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

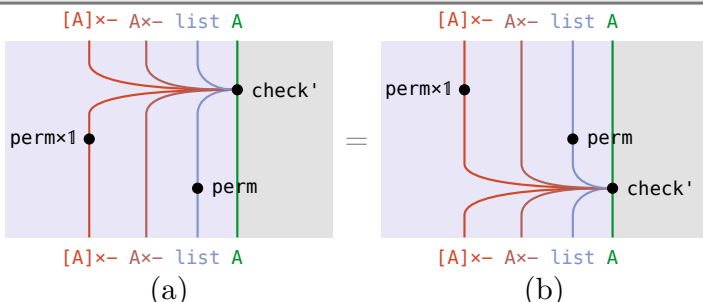
permuting the list  $x \mathrel{++} [a] \mathrel{++} y$  is the same as first permuting  $x$  and  $y$  and then permuting the joined list



perm is transitive and closed under ++

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

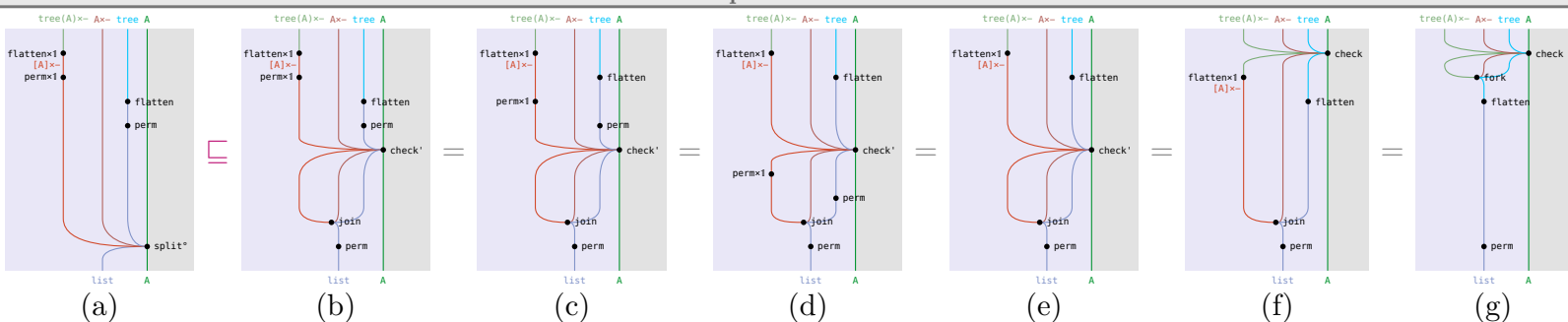


$b \text{ inlist } x \iff b \text{ inlist perm}(x)$

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

**flatten perm**

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



split<sup>°</sup>  $\sqsubseteq$  check' join perm

$(f \times (1 \times f)) (g \times (1 \times g)) = (f g) \times (1 \times (f g))$

(6.6i)

(6.6h)

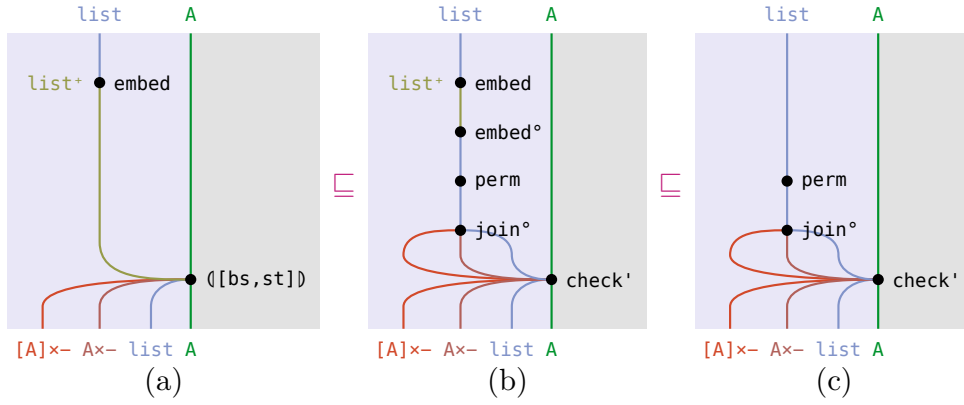
(6.6g)

fork flatten = (flatten × (1 × flatten)) join

$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed}$  (6.6k)

$([bs, st]) \sqsubseteq \text{perm join}^\circ \text{check}'$

turning a list into a non-empty list and folding it with any  $[bs, st]$  meeting the two conditions below gives only **checked triples**  $(x, a, y)$  whose  $\text{join } x \# [a] \# y$  is a permutation of the list

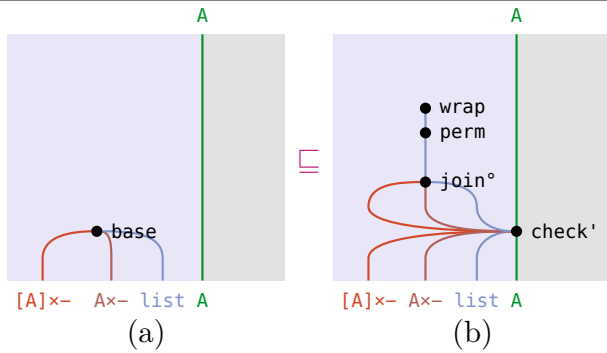


|     |                                                                             |                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{embed } ([bs, st])$                                                  | $bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed } ([bs, st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ |
| (b) | $\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ | $\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$                                                                                                                                                                                    |
| (c) | $\sqsubseteq \text{perm join}^\circ \text{check}'$                          |                                                                                                                                                                                                                                                                                                     |

$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{check}'$

(6.6l)

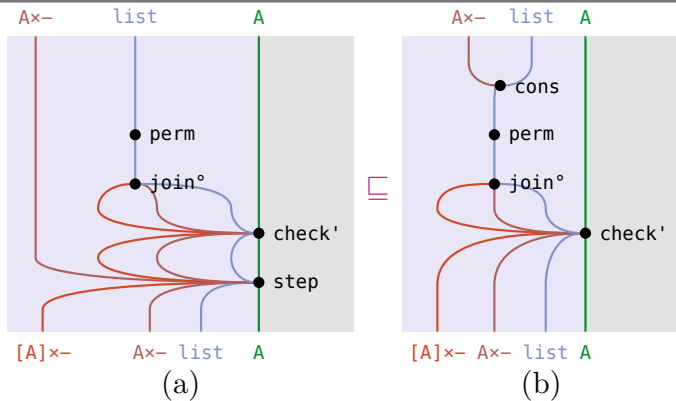
$\text{base}(a)$  is a **checked triple** whose  $\text{join}$  is a permutation of the one-element list  $[a]$



|     |                                                         |                                                                    |
|-----|---------------------------------------------------------|--------------------------------------------------------------------|
| (a) | $\text{base}$                                           | $[] \# [a] \# [] = [a]$ , and $\text{check}'$ holds on empty lists |
| (b) | $\sqsubseteq \text{wrap perm join}^\circ \text{check}'$ |                                                                    |

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{check}'$  (6.6m)

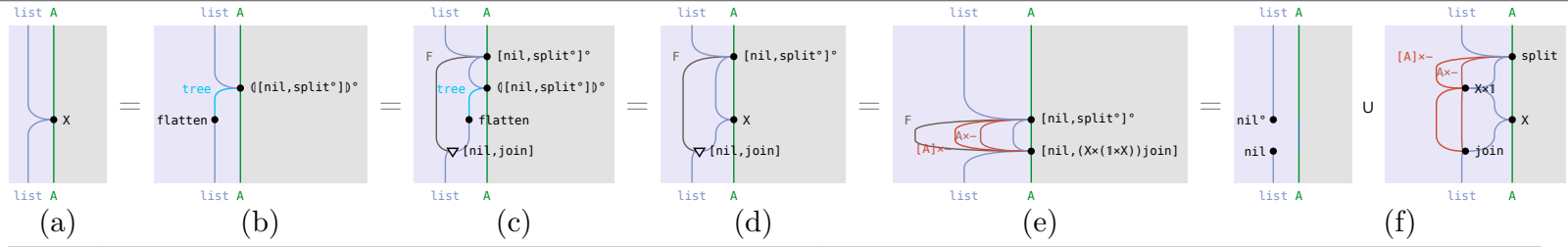
if  $(x, b, y)$  is a **checked triple** for a permutation of  $l$ , then  $\text{step}(a, (x, b, y))$  is a **checked triple** for a permutation of  $[a] \# l$



|     |                                                                  |                                                                               |
|-----|------------------------------------------------------------------|-------------------------------------------------------------------------------|
| (a) | $(1 \times (\text{perm join}^\circ \text{check}')) \text{ step}$ | $aRb$ puts $a$ in front of $x$ ; otherwise $bRa$ and $a$ goes in front of $y$ |
| (b) | $\sqsubseteq \text{cons perm join}^\circ \text{check}'$          |                                                                               |

$$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \Rightarrow X = (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join}) \quad (6.6n)$$

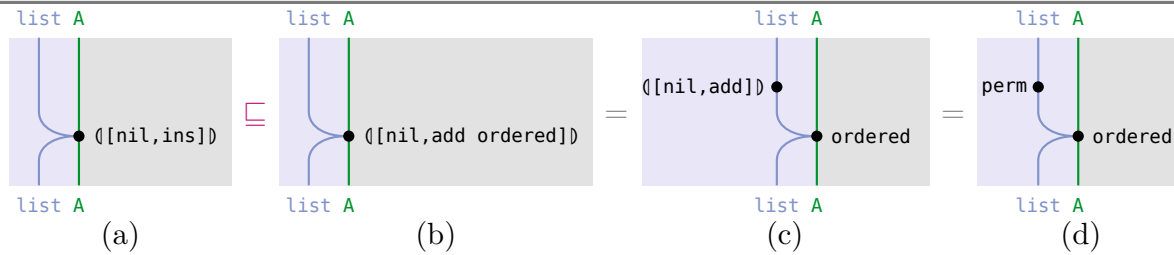
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



|     |                                                                                                                                |                                                                |
|-----|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| (a) | $X$                                                                                                                            | $X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$ |
| (b) | $= ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$                                                                   | $S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$                   |
| (c) | $= [\text{nil}, \text{split}^\circ]^\circ F([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} [\text{nil}, \text{join}]$ | $X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$ |
| (d) | $= [\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$                                                      | $F(S) [T, U] = [T, (S \times (1 \times S)) U]$                 |
| (e) | $= [\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{ join}]$                                  | $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$           |
| (f) | $= (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join})$                                    |                                                                |

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations

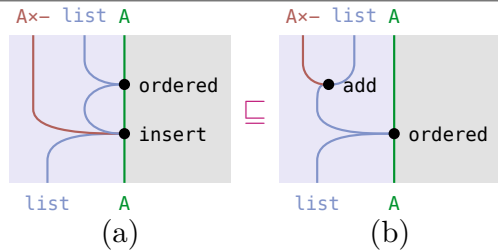


|     |                                                      |                                                                                                     |
|-----|------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{ins}]^\circ$                     | $(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$                              |
| (b) | $\sqsubseteq [\text{nil}, \text{add ordered}]^\circ$ | $\Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq ([\text{nil}, \text{add ordered}])^\circ$ |
| (c) | $= ([\text{nil}, \text{add}])^\circ \text{ ordered}$ | $[\text{nil}, \text{add}]^\circ \text{ ordered} = ([\text{nil}, \text{add ordered}])^\circ$         |
| (d) | $= \text{perm ordered}$                              | $\text{perm} = ([\text{nil}, \text{add}])^\circ$                                                    |

$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \wedge (\forall a. \forall b. \forall c. \quad (6.6p)$$

$$R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered}$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|     |                                            |                                                                        |
|-----|--------------------------------------------|------------------------------------------------------------------------|
| (a) | $(1 \times \text{ordered}) \text{ insert}$ | $\text{insert}$ splices <b>a</b> in, and <b>R</b> transitive and total |
| (b) | $\sqsubseteq \text{add ordered}$           |                                                                        |

## §6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R\***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R\***

$$1 \sqsubseteq R^* \quad (6.7c)$$

**R\*** is reflexive

|     |                              |                          |
|-----|------------------------------|--------------------------|
| (a) | $1$                          | $R \sqsubseteq R \cup S$ |
| (b) | $\sqsubseteq 1 \cup (R R^*)$ | $1 \cup (R R^*) = R^*$   |
| (c) | $= R^*$                      |                          |

$$R \subseteq R^*$$

(6.7d)

 $R^*$  contains  $R$ 

|     |                     |                       |
|-----|---------------------|-----------------------|
| (a) | $R$                 |                       |
| (b) | $= R$               | (6.7c)                |
| (c) | $\sqsubseteq R R^*$ | $R R^* \subseteq R^*$ |
| (d) | $\sqsubseteq R^*$   |                       |

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^*$$

(6.7e)

 $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup R X$ , stated through division

$$R^* R^* \subseteq R^*$$

 $R^*$  is transitive, because  $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup R X$ 

|     |                                    |                                   |
|-----|------------------------------------|-----------------------------------|
| (a) | $(1 \cup (R (R^* / R^*))) R^*$     | $(S \cup T) R = (S R) \cup (T R)$ |
| (b) | $= (R^*) \cup (R (R^* / R^*) R^*)$ |                                   |
| (c) | $= R^* \cup (R (R^* / R^*) R^*)$   | $(R / R) R \subseteq R$           |
| (d) | $\sqsubseteq R^* \cup (R R^*)$     | $R R^* \subseteq R^*$             |
| (e) | $\sqsubseteq R^*$                  |                                   |

$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow 1 \cup (R X) \subseteq X$$

(6.7f)

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $Y \mapsto 1 \cup R Y$ 

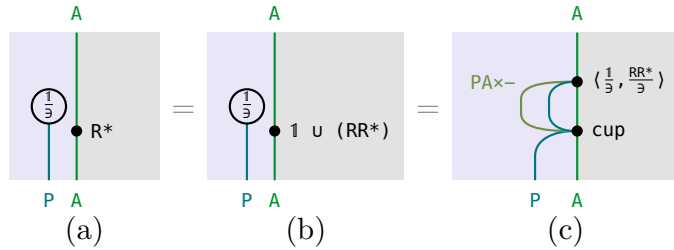
$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow R^* \subseteq X$$

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $X \mapsto 1 \cup R X$ , so it contains  $R^*$ 

|     |                            |                 |
|-----|----------------------------|-----------------|
| (a) | $1 \cup (R X)$             | $1 \subseteq X$ |
| (b) | $\sqsubseteq X \cup (R X)$ | $R \subseteq X$ |
| (c) | $\sqsubseteq X \cup (X X)$ |                 |
| (d) | $\sqsubseteq X$            |                 |

$$\frac{R^*}{\exists} = \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$$

(6.7g)

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors


|     |                                                                         |                                                                                              |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| (a) | $\frac{R^*}{\exists}$                                                   | $1 \cup (R R^*) = R^*$                                                                       |
| (b) | $= \frac{1 \cup (R R^*)}{\exists}$                                      | $\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$ |
| (c) | $= \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$ |                                                                                              |

$$R - \emptyset = R$$

(6.7h)

taking nothing away leaves  $R$ 

$$R \cup (S - R) = R \cup S$$

beside  $R$ , only the part of  $S$  outside  $R$  adds anything

$$R - (S \cup T) = R - S - T$$

taking away  $S$  and then  $T$  takes away  $S \cup T$ 

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking  $T$  away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \Rightarrow (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$$

(6.7i)

the least fixed point of  $\varphi$  after  $\psi$  is  $\varphi$  applied to the least fixed point of  $\psi$  after  $\varphi$ 

$$S R^* = (\mu X : S \cup (X R))$$

(6.7j)

 $S$  followed by any number of  $R$  steps is the least  $X$  containing  $S$  and closed under a further  $R$  step

$$R^* S = (\mu X : S \cup (R X))$$

any number of  $R$  steps followed by  $S$  is the least  $X$  containing  $S$  and closed under an  $R$  step in front

$$\theta(\emptyset, S) = S R^*$$

(6.7k)

started with nothing found and  $S$  to explore,  $\theta$  computes  $SR^*$ 

|     |                                                                  |                                                               |
|-----|------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $\theta(\emptyset, S)$                                           | $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$ |
| (b) | $\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$ | (6.7h)                                                        |
| (c) | $= \emptyset \cup (\mu X : S \cup (X R))$                        | $\emptyset \cup R = R$                                        |
| (d) | $= (\mu X : S \cup (X R))$                                       | (6.7j)                                                        |
| (e) | $= S R^*$                                                        |                                                               |

$$\theta(P, \emptyset) = P$$

(6.7l)

with nothing left to explore,  $\theta$  returns what it has found

|     |                                                          |                                                               |
|-----|----------------------------------------------------------|---------------------------------------------------------------|
| (a) | $\theta(P, \emptyset)$                                   | $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$ |
| (b) | $\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$ | $(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$            |
| (c) | $= P \cup \emptyset$                                     | $R \cup \emptyset = R$                                        |
| (d) | $= P$                                                    |                                                               |